Do Now · retrieval

Floating-point arithmetic and precision

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Normalise positive $0.0100_2 \times 2^4$.

Q2

What is -1 in 8-bit two's complement?

Q3

Calculate 0.5×2^4.

Q4

Calculate binary $0.1000_2 + 0.0100_2$.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Align exponents — Before adding mantissas, both terms need the same power of two. Shift B right one place when raising its exponent by one.

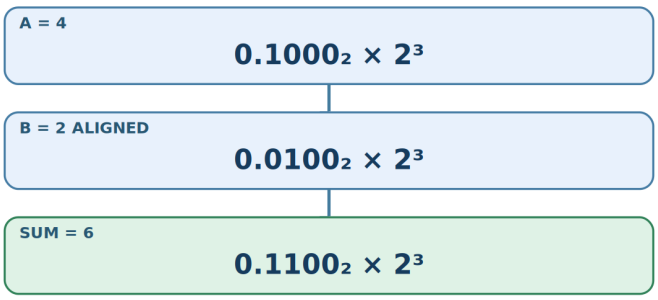
Core idea 2

Add the mantissas — $0.1000_2 + 0.0100_2 = 0.1100_2$; keep the common exponent 3.

Core idea 3

Check in decimal — $0.1100_2 \times 2^3 = 0.75 \times 8 = 6$, matching 4 + 2.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

exponent alignment — Express operands using the same power of two.

rounding — Choose a nearby representable value for an exact result.

truncation — Discard bits beyond the available precision.

overflow — Numerical result exceeds the representable range.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

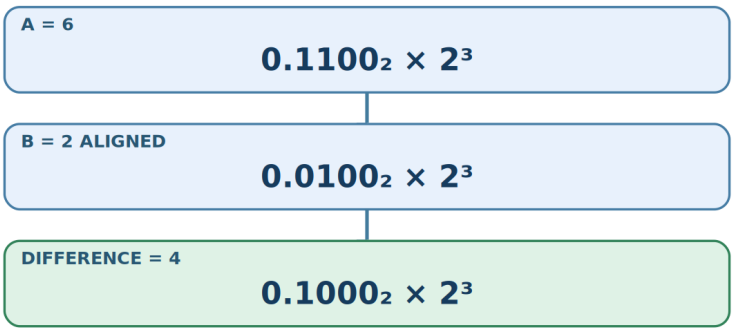
Problem

Calculate $0.1100_2 \times 2^3$ minus $0.1000_2 \times 2^2$.

Follow the worked method

1. Align — B=0.1000×2^2 becomes 0.0100×2^3.
2. Subtract — $0.1100 - 0.0100 = 0.1000$.
3. Check — $0.1000_2 \times 2^3 = 4$; this matches decimal $6 - 2$.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Align
Raise B's exponent 1→3 by shifting its mantissa right twice: 0.1000 becomes 0.0010.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Why must exponents be aligned before binary floating-point addition?

Question 2

Calculate and normalise $0.1010 \times 2^3 - 0.1000 \times 2^2$. Retain all shown mantissa bits during working.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain how exponent alignment can lose precision when the mantissa has a fixed number of bits.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

When increasing an exponent by 1 during alignment, the mantissa must shift right 1 place to preserve its value.

Use key vocabulary from memory

1. Explain exponent alignment in your own words.

2. Explain rounding in your own words.

3. Explain truncation in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.